

Gabriel Cha

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Education

Columbia University

New York, NY

M.S. in Data Science, 3.75 GPA

09/2025 - 12/2026

- Applied Machine Learning, Deep Learning & Neural Networks, Probability & Statistical Inference, Insurance & Banking

University of California, San Diego

San Diego, CA

B.S. in Data Science, 3.75 GPA

09/2021 - 05/2025

- **Teaching Assistant:** Taught data science (*DSC 10, DSC 80*) for 9 quarters to 1,200+ students, holding 120+ office hours
- Data Structure & Algorithms, Statistics & Probability, Data Management, Probabilistic Modeling & Machine Learning

Experience

Root Access

New York, NY

ML Engineering Intern

09/2025 - 05/2026

- Engineered AI agents for firmware developers shipped to enterprise customers, by building a multi-step harness on LangChain/LangGraph with RAG over hardware datasheets and MCU documentation that grounds agent outputs
- Contributed to core agent infrastructure — new tool integrations, sub-agent orchestration, persistent conversational context, CLI user experience improvements, and traceability for output evaluations — deployed on AWS
- Fine-tuned and evaluated open-source LLMs by curating a 9.4k training set from hand-selected GitHub repos, training with QLoRA on Modal A100 GPUs, and benchmarking checkpoints against baseline with a LLM-as-judge evaluations

DSTL Lab

La Jolla, CA

AI/ML Research Assistant

10/2024 - 05/2025

- Co-authored paper accepted @ SIGCSE 2026 on a two-stage context engineering harness that retrieves and summarizes raw content into structured LLM inputs, improving generation accuracy from 23% to 87% on 297 outputs
- Created a quantitative evaluation framework with 3 domain-specific metrics and curated 91 test cases to benchmark LLM output quality, driving iterative improvements across prompt design and context window optimization

Edison

Rosemead, CA

Data Science Intern

06/2024 - 09/2024

- Engineered scalable system monitoring 70+ solar plants for automated outlier detection, transforming 3-day manual processes into 10-minute automated workflows preventing \$5K+ monthly losses from undetected system errors
- Developed ETL pipeline for third-party data requests, processing large datasets (1TB+) from 100+ files and eliminating 4-day manual reporting cycles to accelerate data delivery for clients in the energy trading industry

Projects

Cloi - Debugging AI Agent [\[395 Github Stars\]](#)

05/2025

- Developed AI debugging agent that analyzes error messages, retrieves contextually relevant code files via RAG, and applies LLM generated chain-of-thought fixes to codebase. *[Invited to Y Combinator Batch S25 for Interview]*

Classifying Emotions with CNNs

09/2023

- Led \$6,500 UCSD-funded research project using ResNet-50 architecture for speech emotion classification; implemented augmentation pipeline and trained CNN on PyTorch, achieving 86% accuracy on test datasets

Skills

Programming Languages

Python, Jupyter Notebook, Git, SQL, Java, Typescript

AI Agents

LangChain, LangGraph, LLMs, Embeddings (RAG), Chunking, Context Engineering

Machine Learning

TensorFlow, PyTorch, Scikit-learn, Hugging Face, Natural Language Processing (NLP)

Data Analytics

Pandas, Spark, Numpy, Tableau, Data Cleaning, A/B Testing, Statistical Analysis

Soft Skills

Curiosity-driven with strong attention to detail, Collaborative cross-functional contributor